

Lab 5: TCP Analysis with Wireshark

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Overview

We have covered many TCP details and practiced connecting those ideas to what you see in Wireshark when analyzing a `.pcap` file.

In this lab, you will apply your TCP and Wireshark knowledge to analyze the file `mystery.pcap`.

Write-Up

Analyze the pcap file and write a report that answers all of the questions below

You must include **supporting evidence** for every claim. For example, if you answer a question about ports, retransmissions, or file contents, you should point to specific packets, fields, timing, sequence numbers, or Wireshark features that support your answer.

Your goal is not just to give an answer, but to explain **how you determined it**.

Questions

1. **What port numbers and IP addresses are used in this pcap?**

Be sure to identify all relevant endpoints involved in the TCP conversation.

Source IP: 127.0.0.1

Destination IP: 127.0.0.1

Source Port: 43060

Destination port: 5432

The IP addresses were found directly in the information columns, I found the port information under the first two rows in TCP window.

2. **Who is the client? Who is the server? Explain how you know this.**

Use the TCP connection setup to justify your answer.

Source (client) Port: 43060

Destination (server) port: 5432

The client sends the 3 way handshake by sending the syn packet 1 and the server responds with the ack being packet 2.

3. **This capture appears to show a local connection between processes on the same machine, rather than traffic between two different physical hosts.**

Briefly describe the evidence in the pcap that supports this conclusion.

I think the most obvious evidence this is a local connection is the src and dst IP packets are on 127.0.0.1 instead of different addresses which shows they're communicating on a local connection. Another thing I noticed is how quickly these packets are communicating which indicated how short of a distance they have to go to communicate.

No.	Time	Source	Destination
1	0.000000	127.0.0.1	127.0.0.1
2	0.000010	127.0.0.1	127.0.0.1
3	1.018418	127.0.0.1	127.0.0.1
4	1.018427	127.0.0.1	127.0.0.1
5	1.018456	127.0.0.1	127.0.0.1
6	1.018461	127.0.0.1	127.0.0.1
7	1.018476	127.0.0.1	127.0.0.1
8	1.018660	127.0.0.1	127.0.0.1
9	1.018666	127.0.0.1	127.0.0.1
10	1.018681	127.0.0.1	127.0.0.1
11	1.018685	127.0.0.1	127.0.0.1

4. **What is the MSS for the client and server?**

State where you found this information in the TCP connection setup.

Client MSS: 65495 bytes

Server MSS: 65495 bytes

This was under the selected packets TCP option information.

5. **Are any packets larger than the MSS? If so, which ones?**

Compare actual payload sizes to the advertised MSS values.

I could not find any packets larger than the negotiated MSS.

6. **What causes retransmissions in general?**

Answer this as a TCP concept question. Explain what typically causes TCP to resend data.

In TCP, things like packet loss due to network congestion for example, lost or delayed acknowledgments, or out of order packets are typical reasons why TCP resends data.

7. **What might have caused the retransmissions in this pcap? Provide evidence.**

Hints:

1. Look at the **timing** between the original packet and the retransmitted one.
2. Look for repeated **sequence numbers**.
3. Check whether acknowledgments are delayed, missing, or repeated.
Use the packet timing and sequence/acknowledgment behavior to support your answer.

4	1.018427	127.0.0.1	127.0.0.1	TCP	74 [TCP Retransmission] 5432 → 43060 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=
5	1.018456	127.0.0.1	127.0.0.1	TCP	74 [TCP Retransmission] 5432 → 43060 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=

Duplicate packets (sequence numbers)

What might've caused the retransmission is the timeout since the client and server expired both retransmitted them.

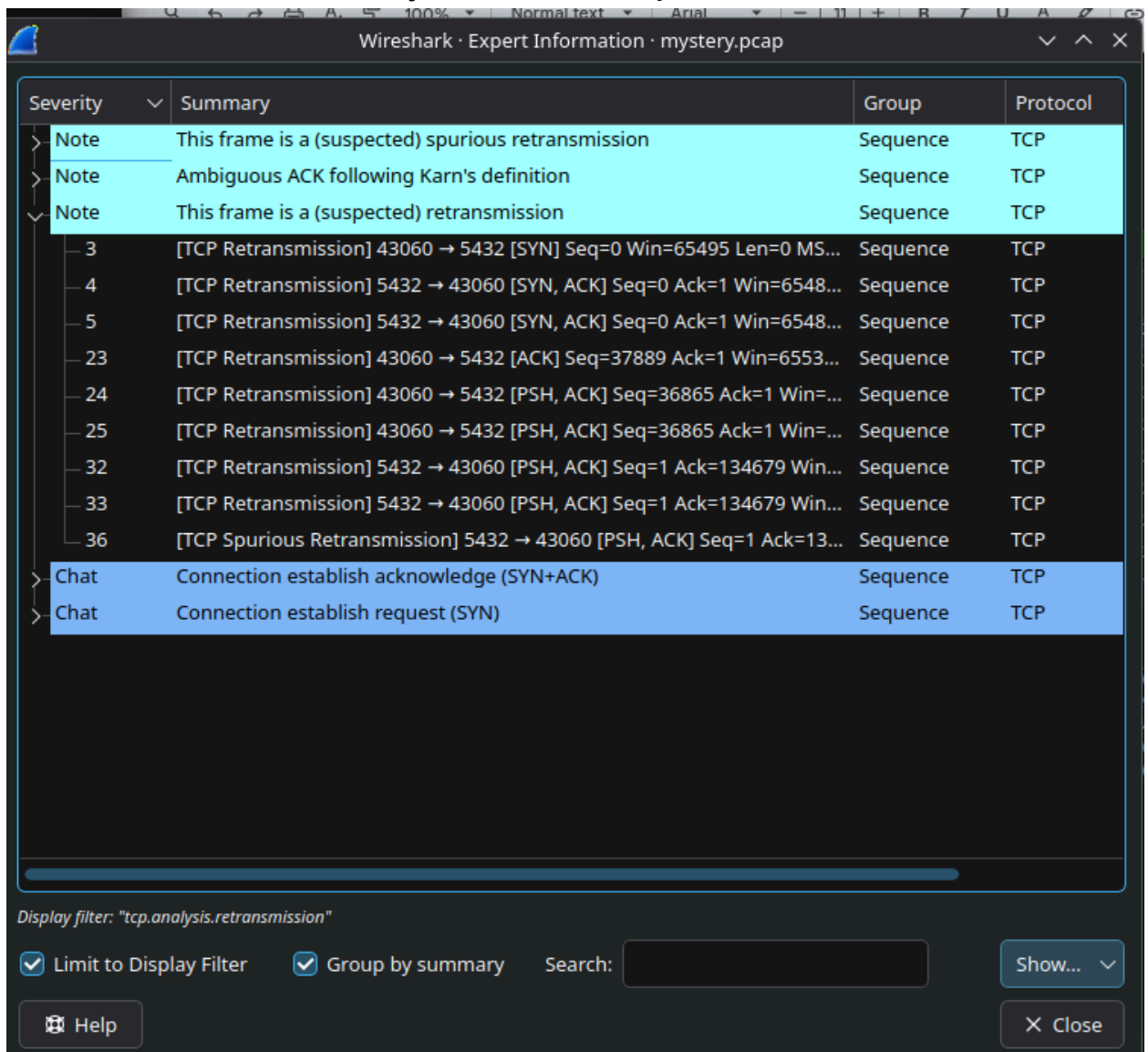
8. **What was the retransmission rate?**

Hint: A simple way to calculate this is: $\text{Retransmission Rate} = \frac{\text{Number of retransmitted packets}}{\text{Total number of packets}}$

Clearly state what you counted and how you got your final percentage.

There are 39 total packets and 9 retransmitted packets so the retransmission rate is $\frac{9}{39} = .2376 * 100 = 23.76\%$

9. What Wireshark filter shows only the retransmitted packets?



The image shows the Wireshark Expert Information window for a file named 'mystery.pcap'. The window displays a list of packets with columns for Severity, Summary, Group, and Protocol. The first three packets are highlighted in light blue and are all 'Note' severity. The first packet is 'This frame is a (suspected) spurious retransmission', the second is 'Ambiguous ACK following Karn's definition', and the third is 'This frame is a (suspected) retransmission'. The remaining packets are 'TCP Retransmission' or 'TCP Spurious Retransmission' for various sequence numbers and flags. The last two packets are 'Chat' severity, showing 'Connection establish acknowledge (SYN+ACK)' and 'Connection establish request (SYN)'. At the bottom, the display filter is set to 'tcp.analysis.retransmission', and there are checkboxes for 'Limit to Display Filter' and 'Group by summary', both of which are checked. There is also a search bar and buttons for 'Show...', 'Help', and 'Close'.

Severity	Summary	Group	Protocol
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP
Note	Ambiguous ACK following Karn's definition	Sequence	TCP
Note	This frame is a (suspected) retransmission	Sequence	TCP
	[TCP Retransmission] 43060 → 5432 [SYN] Seq=0 Win=65495 Len=0 MS...	Sequence	TCP
	[TCP Retransmission] 5432 → 43060 [SYN, ACK] Seq=0 Ack=1 Win=6548...	Sequence	TCP
	[TCP Retransmission] 5432 → 43060 [SYN, ACK] Seq=0 Ack=1 Win=6548...	Sequence	TCP
	[TCP Retransmission] 43060 → 5432 [ACK] Seq=37889 Ack=1 Win=6553...	Sequence	TCP
	[TCP Retransmission] 43060 → 5432 [PSH, ACK] Seq=36865 Ack=1 Win=...	Sequence	TCP
	[TCP Retransmission] 43060 → 5432 [PSH, ACK] Seq=36865 Ack=1 Win=...	Sequence	TCP
	[TCP Retransmission] 5432 → 43060 [PSH, ACK] Seq=1 Ack=134679 Win...	Sequence	TCP
	[TCP Retransmission] 5432 → 43060 [PSH, ACK] Seq=1 Ack=134679 Win...	Sequence	TCP
	[TCP Spurious Retransmission] 5432 → 43060 [PSH, ACK] Seq=1 Ack=13...	Sequence	TCP
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP
Chat	Connection establish request (SYN)	Sequence	TCP

Display filter: "tcp.analysis.retransmission"

☒ Limit to Display Filter ☒ Group by summary Search: Show...

If you press analyze and expert information, this window will appear and you can find the retransmitted packets. You can also type `tcp.analysis.retransmission` at the search bar in the main interface of wire shark and it will pull the retransmitted packets.

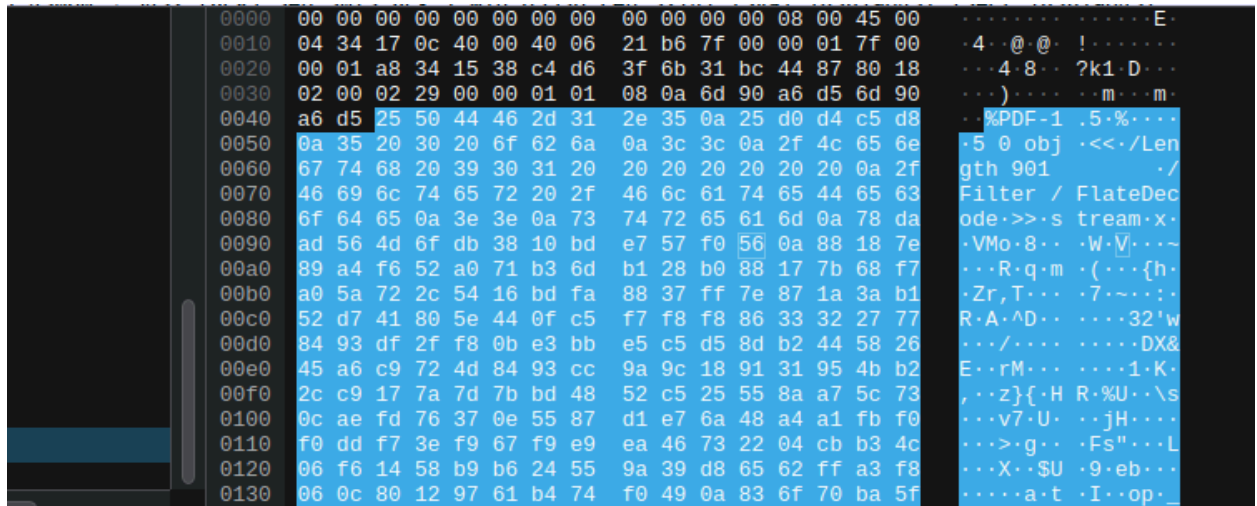
10. Does the packet loss appear to affect the transmission rate?

Use evidence from timing, packet spacing, or pauses in the transfer to support your answer.

Yes, it definitely slows it down. At the beginning there is a 1 second delay which causes the first ack to expire before it can try again. After TCP saw the retransmission, it assumed the network was clogged and shrunk the amount of data being sent at once which makes the whole transfer take much longer than it should.

11. Can you tell what file type was sent? If so, what was it?

Hint: Inspect the beginning of the transmitted application data. Many file types have recognizable “magic bytes” or file signatures.



The signature is %PDF-1.5%

12. You reconstruct the sent data from the pcap. To reconstruct the data, follow these steps:

1. Identify a packet that belongs to the TCP stream carrying the file.
2. In Wireshark, right-click the packet and choose **Follow** → **TCP Stream**.
3. In the stream window, select the correct **direction** of the transfer.
4. Change the display format to **Raw**.
5. Save the stream data to a file.
6. Open the saved file and determine what was sent.

After doing this, state what the transferred file or content was.

CSCD 330-040 Computer Networks

Lab 1, tool scavenger hunt - Due, September 27, 2023

September 20, 2023

Instructions:

The command line tool `man` is your friend. Use `man` and the Internet to answer the following questions. Don't copy the explanation of `man` for your answers, but use it as a guide to run the tool and find out what it's doing.

Questions:

1. What does `ping` do?
2. How could `ping` be useful?
3. What does `dig` do?
4. What does `nslookup` do?
5. Do they tell you any of the same information? If so, what?
6. What is a domain name?
7. What is a DNS server?
8. What DNS server does your machine use?
9. How would you change this on a Linux machine?
10. What is `Wireshark`?
11. What is `traceroute`?
12. What is the difference between running it with `-T` vs `-I`?
13. What is `whois`?
14. What is RFC 3192?
15. On the topic of RFCs, what is an RFC?
16. Use `nslookup` and `dig` on the same domain (e.g., `google.com`). Do they return the same IP address. What happens when you put the IP address(es) in the address bar of your browser?
17. What does `ip a` output?
18. What is `tcpdump`?
19. Is there anything specific you want to learn about networks? If so, what? If not, what do you hope to get out of this class?
20. Tell me something about yourself.

Lab 1 tool scavenger hunt and all the content seen in the photo above.

13. Do you see a DNS call? If so, what port? If not, why not?

Explain your reasoning using the protocols and endpoints visible in the capture.

There is no DNS call in this capture because the connection is local. DNS isn't even needed because the computer is talking to itself and doesn't need to find an IP address especially since both endpoints are 127.0.0.1 rendering DNS server unnecessary.

Include relevant screenshots or figures if they help support your analysis.

Citations

If you use outside sources, cite them appropriately. You may also cite the textbook or class slides when making general TCP-related claims.